

PEDRAM MOHAMMADI

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PROFESSIONAL SUMMARY

- Expert in algorithm development for image/video processing, specializing in subjective and objective quality assessment, computational photography, color processing, and video encoding.
- Proficient in Python, MATLAB, C, and C++, with hands-on experience in OpenCV, Scikit-learn, and FFMPEG.
- Proven leader with a track record of delivering patented innovations and mentoring teams to drive high-impact solutions in image/video processing technologies.

EDUCATION

Ph.D. in Electrical and Computer Engineering

2015 - 2020

University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering

2012 - 2014

Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer

2021 - Present

NETINT Technologies Inc., Canada

- Identified and recommended optimal IP vendors for acquisition, enhancing product capabilities and supporting company growth.
- Led the development of patented algorithms to optimize resource allocation, block-level processing, and prediction enhancement, achieving up to 12% improvement in efficiency metrics across multiple platforms.
- Designed a patented real-time conversion suite for dynamic range adjustments (SDR-to-HDR, HDR-to-SDR, HDR-to-HDR), meeting customer requirements and stringent quality standards.
- Conducted comprehensive quality evaluations and fine-tuning, set benchmarks, and achieved an average 5% increase in objective video quality across various product lines.
- Mentored junior team members and collaborated with hardware engineers to ensure seamless algorithm implementation and integration.

Scientific Research and Experimental Development (SR&ED) Consultant

2020 - 2021

KPMG LLP, Canada

- Evaluated technical R&D projects across diverse industries to determine eligibility for CRA SR&ED tax incentives, leveraging strong analytical skills to assess complex technical and financial data.
- Collaborated with clients to maximize tax credit claims, achieving 20-30% annual increases (totaling 500K to 2M in credits), allowing significant reinvestment into innovative R&D initiatives.

- Managed multiple client engagements, demonstrating effective project management and prioritization under tight deadlines in a fast-paced consulting environment.

Application Developer I

2018 - 2019

TELUS Communications Inc., Canada

- Led a team of two engineers to implement a real-time SDR-to-HDR conversion system based on my PhD research, achieving real-time performance for 1080p content.
- Optimized the algorithm for real-time execution while maintaining high visual quality, verified through comprehensive objective and subjective video quality assessments.
- Collaborated with executive stakeholders (including CTO) to define project requirements and ensure alignment with business goals, resulting in successful deployment in the TELUS content services pipeline, which now improves video quality for subscribers.

Research Scientist

2015 - 2020

The University of British Columbia, Canada

- Led research on SDR-to-HDR conversion, developing a patented algorithm that improved subjective visual quality by 80% over state-of-the-art methods, validated through rigorous subjective and objective evaluations.
- Innovated supporting algorithms for adaptive brightness segmentation using entropy theory, flickering reduction, and preservation of color hue, achieving 70% better color accuracy.
- Published multiple papers in prestigious IEEE journals and conferences, contributing to the advancement of the field.

PATENTS

- **P. Mohammadi**, J. Li, E. Andrade Neto, and J. Lou "Methods and Apparatus for video rate control," Patent filed, filing date: January 2024.
- **P. Mohammadi**, E. Andrade Neto, and J. Lou "Image processing methods, devices, electronic devices and storage media," China Patent Office, Application No. CN116167950B, Publication date: April 2023.
- P. Nasiopoulos, S. Ploumis, M. T. Pourazad, R. Boitar, and **P. Mohammadi**, "Methods and Apparatuses for Tone Mapping and Inverse Tone Mapping," U.S. Patent, Application No. US11100888B2, Publication date: January 2019.

PUBLICATIONS

For a complete list of my publications please click [here](#)

WORK AUTHORIZATION

Currently holding a Canadian citizenship